

Perceiver & Perceiver IO

Implementation and Experimental Analysis

Francesco Gigli e Corso Vignoli

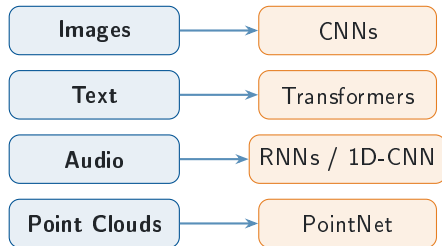
Università degli Studi di Firenze
Course B031278 — Deep Learning

Agenda

- 1 Introduction & Motivation
- 2 Perceiver
- 3 Perceiver IO
- 4 Discussion

The Problem: Modality-Specific Architectures

One modality, one architecture



Each data type is usually paired with a dedicated model and domain-specific assumptions about structure.

The scaling bottleneck

Self-attention cost

$$\mathcal{O}(M^2 \cdot d)$$

grows quadratically with input length M

Input representation	M
16 × 16 image patches	256
raw 224 × 224 pixels	50,176

Raw pixels make standard attention
impractical.

Core question: can one architecture handle arbitrary inputs without quadratic cost?

The Perceiver Solution

Key idea (Jaegle et al., ICML 2021):

- 1 Compact **latent array**, $N \ll M$
- 2 **Cross-attention**: latents query the input
- 3 **Latent self-attention** refines latents
- 4 Iterate with **weight sharing**

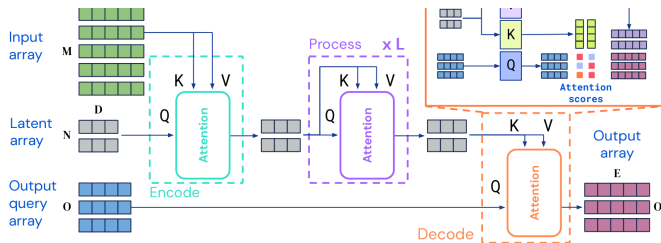
Component	Complexity
Cross-Attention	$\mathcal{O}(M \cdot N)$
Latent Self-Attn	$\mathcal{O}(N^2)$
Total	$\mathcal{O}(M \cdot N + N^2 \cdot L)$

Since $N \ll M \Rightarrow$ linear in M !

Project contribution:

- *From-scratch* PyTorch: Perceiver + IO
- 3 modalities: images, point clouds, text
- Complete ablation study

Perceiver IO: Structured Outputs



Key extension (Jaegle et al., ICLR 2022):

- **Encode:** input $\rightarrow$ latent (cross-attn)
- **Process:** latent self-attention ($\times L$)
- **Decode:** **output queries** read latents

Flexibility:

- 1 query $\rightarrow$ classification
- O queries $\rightarrow$ structured output (MLM)
- $H \times W$ queries $\rightarrow$ optical flow

Architectural diagram

- **Input Array** ($M \times C$): flat generic sequence (pixels, 3D points, bytes)
- **Latent Array** ($N \times D$): learned compact bottleneck, $N \ll M$
- **Weight Sharing**: cross-attn + self-attn block reused across iterations
- **Output**: mean pooling over latents $\rightarrow$ classification head

Cross-Attention: The Key Mechanism

Formally:

$$Q = X_{\text{latent}} W_Q \in \mathbb{R}^{N \times d_k}$$

$$K = X_{\text{input}} W_K \in \mathbb{R}^{M \times d_k}$$

$$V = X_{\text{input}} W_V \in \mathbb{R}^{M \times d_v}$$

$$\text{CrossAttn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

The **latents are the queries**, the input provides keys and values $\Rightarrow$ informational *bottleneck*.

Architectural diagram

Key insight:

- Attention matrix: $N \times M$ (not $M \times M$)
- Complexity: $\mathcal{O}(N \cdot M \cdot d_k)$
- Decouples depth from input size

Latent Transformer & Weight Sharing

Self-Attention among latents:

$$Q = K = V = X_{\text{latent}} \Rightarrow \mathcal{O}(N^2)$$

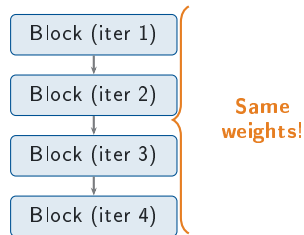
- Pre-norm LayerNorm
- Residual connections
- Feed-forward with $4\times$ expansion
- GEGLU activation (gated linear units)

GEGLU:

$$\text{GEGLU}(x) = (xW_1) \odot \text{GELU}(xW_2)$$

Higher non-linear capacity, stable gradient flow.

Weight sharing mechanism:



Configuration	Parameters
Unshared ($T=4$ iters)	$\sim 8.67\text{M}$
Shared (T iterations)	$\sim 3.35\text{M}$
Reduction	$\sim 2.6\times$

Fourier Positional Encoding

Why is it needed?

- Attention is *permutation-equivariant*
- Without PE the model has no notion of “where” data is
- Experimentally confirmed: **61.34% without PE** vs 72.02% with PE

Fourier PE (per axis x_d , raw coord. concatenated):

$$\gamma(x_d) = [x_d, \sin(f_k \pi x_d), \cos(f_k \pi x_d)]_{k=1}^K$$

$K = 64$ frequencies **linearly spaced** in $[1, \mu/2]$ ($\mu/2 = \text{Nyquist}$).

Resulting dimensions:

Modality	Pos. features	C_{tot}
Image (2D)	$3 + 2(2K+1)$	261
Points (3D)	$3 + 3(2K+1)$	390
Text (byte)	$128 + 64$	192

Critical Finding

Positional encoding is the **single most important** component: removing it costs **−10.68pp** (72.02% $\rightarrow$ 61.34%).

Architectural diagram

Decoder cross-attention (single layer):

$$Q_{\text{dec}} = \text{OutputQuery } W_Q, \quad K_{\text{dec}} = \text{Latent } W_K, \quad V_{\text{dec}} = \text{Latent } W_V$$

$$\text{Output} = \text{softmax}\left(\frac{Q_{\text{dec}} K_{\text{dec}}^T}{\sqrt{d_k}}\right) V_{\text{dec}} \in \mathbb{R}^{O \times E}$$

- The number of output queries O is **task-dependent** and independent of N and M
- A single decoder cross-attention layer suffices (no self-attention among outputs)

Output Query Design per Task

Task	O (queries)	Query Design	Output
Classification	1	Single learned vector	Class logits
Optical Flow	$H \times W$	Position encodings	Per-pixel flow
Masked LM	N_{masked}	Position of masked tokens	Token logits (vocab=256)
Multimodal	Variable	Task-specific embeddings	Mixed outputs

Our Classification

Single query $\rightarrow$ linear projection $\rightarrow C$ classes.
Used for CIFAR-10 and ModelNet40.

Our MLM

One query per masked position $\rightarrow$ project to vocab size 256. Byte-level prediction on WikiText-103.

Computational Complexity Analysis

Model	Self-Attn	Cross-Attn	Total
Standard Transformer	$\mathcal{O}(M^2 d)$	—	$\mathcal{O}(M^2 d \cdot L)$
Perceiver	$\mathcal{O}(N^2 d)$	$\mathcal{O}(MNd)$	$\mathcal{O}((MN + N^2 L) d)$
Perceiver IO	$\mathcal{O}(N^2 d)$	$\mathcal{O}((M+O)Nd)$	$\mathcal{O}((MN + N^2 L + ON) d)$

Concrete example (CIFAR-10):

- $M = 1024$ (pixels, 32×32), $N = 96$, $L = 4$
- Standard: $1024^2 = 1,048,576$
- Perceiver:
 $1024 \times 96 + 96^2 \times 4 = 135,168$
- $\sim 7.8 \times$ reduction

Scaling advantage:

- For images 224×224 : $M = 50k$
- Transformer: 2.5×10^9 ops
- Perceiver ($N = 512$): $\sim 26M$ ops
- $\sim 100 \times$ reduction

GELU Activation & LAMB Optimizer

GELU (Gaussian Error Linear Unit):

$$\text{GELU}(x) = x \cdot \Phi(x) \approx x \cdot \sigma(1.702 x)$$

- Smooth, non-monotonic activation
- Standard in Transformer architectures
- Allows small negative gradient flow

LAMB (Layer-wise Adaptive Moments):

$$r_t^{(l)} = \frac{\|w_t^{(l)}\|}{\|w_t^{(l)} + \eta \hat{u}_t^{(l)}\|}$$

- Per-layer **trust ratio** $r_t^{(l)}$

Why LAMB for Perceiver?

- Cross-attention bottleneck creates high variance gradients
- Standard Adam **destabilizes** with large batches
- LAMB safely scales batch sizes up to 1024
- Critical for the multi-scale latent transformer

Optimizer	Large Batches	Layer-wise LR
Adam	Diverges	Global
LAMB	Stable	Per-layer trust

Implementation: Code Structure

Project organization:

```
Perceiver_project/  
  perceiver/  
    model.py  
    perceiver_io.py  
    attention.py  
    positional_encoding.py  
    lamb_optimizer.py  
  data/  
    cifar10_datamodule.py  
    modelnet40_datamodule.py  
    wikitext_datamodule.py  
    glue_datamodule.py  
  experiments/  
    run_cifar10.py  
    run_modelnet40.py  
    run_wikitext.py
```

Key numbers:

Component	Lines
Core model	~600
Data modules	~500
Experiments	~800
Utilities & analysis	~600
Total	~2500

Design choices:

- PyTorch + PyTorch Lightning
- Modular architecture
- All experiments reproducible via single script

CIFAR-10: Experimental Setup

Goal: validate the Perceiver on raw pixels without any CNN assumptions.

Hyperparameter	Value
Latent dim ($N \times D$)	96×384
Cross-attn stages	4
Transformer blocks	4 (shared)
Attention heads	3
Patch size	2×2
Dropout	0.2
Optimizer	LAMB
Learning rate	0.004
Batch size	64
Epochs	120
Total parameters	$\sim 3.35\text{M}$

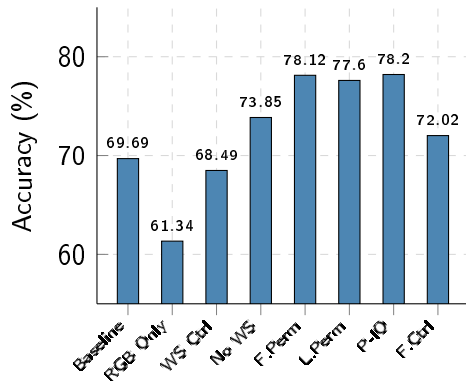
Ablation study plan:

- ① Impact of positional encoding
- ② Weight sharing vs no sharing
- ③ Robustness to pixel permutation
- ④ Perceiver vs Perceiver IO

Data pipeline:

- 32×32 RGB images, pixel-level input ($M=1024$)
- 2D Fourier positional encoding ($K=64$)
- Standard augmentation: horizontal flip, random crop

CIFAR-10: Ablation Results

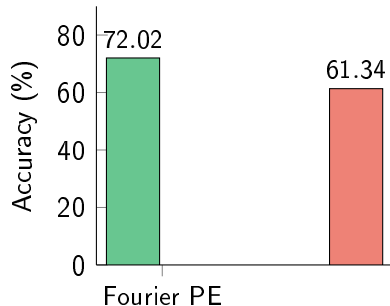


Experiment	Acc.
Perceiver IO	78.20%
Fourier Permuted	78.12%
Learned PE Permuted	77.60%
No Weight Sharing	73.85%
Fourier Control	72.02%
Baseline Fourier	69.69%
WS Control	68.49%
RGB Only (no PE)	61.34%

Key observations:

- PE is **essential**: -10.68pp without it
- Permuted-pixel runs are **best** (78.12%), not worst
- Weight sharing: -5.36pp but $\sim 2.6\times$ fewer params

CIFAR-10: Impact of Positional Encoding



Why does this happen?

- Attention is *permutation-equivariant*
- Without PE the model cannot exploit spatial positions
- Color-only signal still classifies, but clearly weaker

Quantitative impact:

- With Fourier PE: **72.02%**
- Without PE (RGB only): **61.34%**
- Delta: **−10.68 percentage points**
- Still well above chance (10%), but markedly worse

This confirms the original paper's
Perceiver & Perceiver IO

CIFAR-10: Weight Sharing Analysis

Comparison:

Config	Params	Acc.
Shared (baseline)	3.35M	69.69%
WS Control	3.35M	68.49%
No Sharing	8.67M	73.85%

Trade-off:

- No sharing: +5.36pp accuracy
- But $\sim 2.6\times$ more parameters
- Efficiency ratio: ~ 1.0 pp per 1M extra params

Why weight sharing works:

- Acts like a *recurrent* processor in latent space
- Iterative refinement of representations
- Implicit regularization effect
- Critical for deployment on limited hardware

Practical Implication

Weight sharing reaches **92.7%** of the no-sharing accuracy with only **39%** of the parameters — a strong efficiency trade-off for resource-constrained scenarios.

CIFAR-10: Permutation Robustness

Experiment: train *and* evaluate on pixels shuffled by a single fixed permutation σ (same σ throughout).

Configuration	Acc.
Fourier PE (natural order)	69.69%
Fourier PE (permuted)	78.12%
Learned PE (permuted)	77.60%

Counterintuitive result:

- Permuting pixels **raises** accuracy: +8.43pp
- Fourier vs learned (permuted): +0.52pp

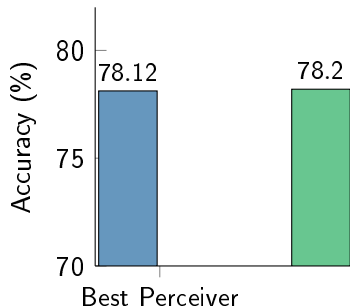
Analysis:

- The Perceiver has **no built-in 2D bias**: position is read only from the PE
- With a fixed σ in train *and* test, it learns the layout from the Fourier features alone
- **Fourier PE** (deterministic) slightly beats learned PE and generalizes better

Insight

Permutation invariance is a **feature**, not a weakness: the model assumes nothing about input ordering, exactly as the paper argues (Jaegle et al., 2021, Tab. 2).

CIFAR-10: Perceiver vs Perceiver IO



Why the decoder helps:

- Output query is a *learned aggregation*, strictly more expressive than mean pooling (paper Fig. 6)
- Decoder cross-attention attends selectively to relevant latents
- Small but consistent gain, with smoother convergence
- Cost: $\sim 3\times$ params (9.5M vs 3.35M)

Model	Acc.	Δ
Best Perceiver (exp6)	78.12%	—
Perceiver IO	78.20%	+0.08pp

Configuration:

Hyperparameter	Value
Latent dim ($N \times D$)	128×512
Cross-attn stages	2
Transformer blocks	6 (shared)
Points per object	2048
Optimizer	LAMB
Epochs	200
Batch size	128

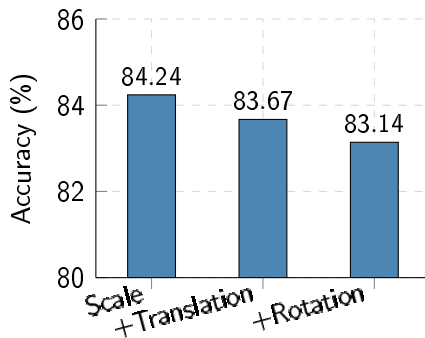
Results:

Configuration	Acc.
Baseline (scale)	84.24%
+ Translation	83.67%
+ Rotation	83.14%
<i>Original Paper</i>	<i>85.7%</i>

Excellent Reproduction

Gap of only **1.46pp** vs the original paper, despite using batch size 128 vs 512 (GPU constraints).

ModelNet40: Augmentation Effects



Analysis:

- **Scale only** (baseline): best result at 84.24%
- Adding translation: -0.57pp
- Adding rotation: -1.10pp

Why augmentation doesn't help:

- With 3D Fourier PE the Perceiver already handles rotations natively
- So rotation augmentation is *redundant* — adds noise, not signal
- ModelNet40 objects are already centred, limiting translation's effect

Comparison with paper (85.7%):

WikiText-103: Masked Language Model Pre-training

Goal: pre-train a general language encoder without specialized tokenizers.

Component	Detail
Tokenization	Byte-level (UTF-8)
Vocabulary size	256
Sequence length	1024
Mask probability	15%
Model	Perceiver IO ($\sim 10M$)
Epochs / batch	50 / 32
Masked byte acc.	82.20%

Why byte-level?

- No BPE/WordPiece tokenizer needed
- Truly language-agnostic input
- Fixed vocabulary of size 256
- Trade-off: longer sequences for same text

MLM task:

- 15% of byte positions masked
- Output queries = masked positions
- Decoder predicts original byte values
- Cross-entropy loss over 256 classes

GLUE: Fine-tuning on 8 Tasks

Transfer learning pipeline:

Architectural diagram

Fine-tuning setup:

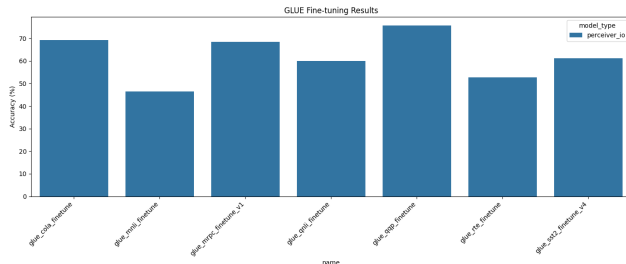
- Transfer pre-trained MLM encoder
- Replace output queries with classification query
- 30 epochs, $LR = 5 \times 10^{-4}$
- Byte-level input (same as pre-training)

8 GLUE benchmark tasks:

- **Single sentence:** CoLA, SST-2
- **Similarity:** MRPC, STS-B, QQP
- **Inference:** MNLI, QNLI, RTE

Each task tests a different NLU capability with the same architecture.

GLUE: Per-Task Analysis



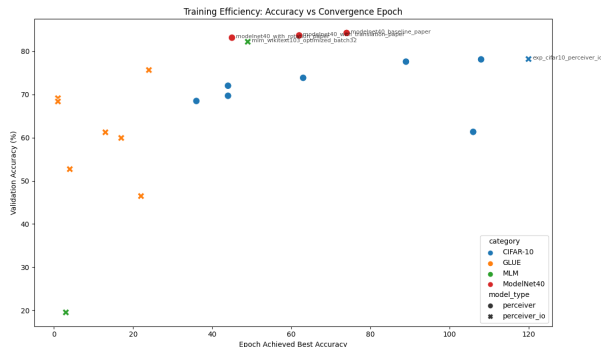
Key observations:

- **QQP** (paraphrase): best task at 75.65%
- **CoLA** 69.13%, **MRPC** 68.38% follow
- **MNLI** (46.47%) & **RTE** (52.71%): hardest — need deep NLI reasoning
- Mean $\sim 63\%$ vs paper $\sim 81\%$

Limitations:

- Byte-level tokenization produces long sequences
- Smaller model ($\sim 10M$) vs paper's IO (201M)

Convergence: All Experiments



Convergence speed by modality:

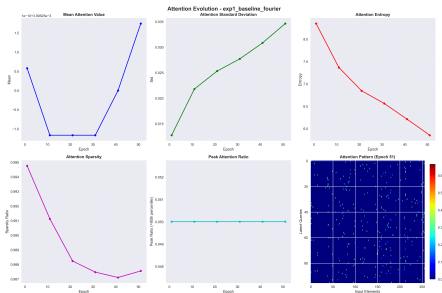
Modality	Epochs
Point Clouds	~50 (fast)
Text (GLUE FT)	~15–30
Images (CIFAR-10)	~120 (slow)

Observations:

- Point clouds: low-dimensional input, fast convergence
- GLUE: transfer learning accelerates training
- CIFAR-10: pixel-level input requires many iterations for spatial learning

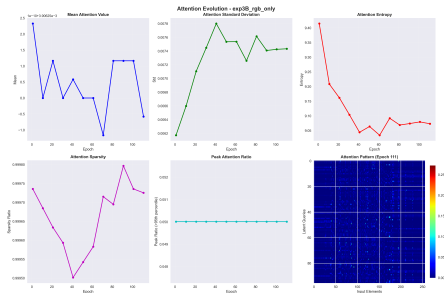
Attention Maps Analysis

Fourier PE (Epoch 81)



Structured, spatially selective patterns (entropy ~ 6.4). Latents specialize in different image regions.

RGB Only — no PE (Epoch 81)



Diffuse, near-uniform patterns (entropy ~ 9.1). No spatial selectivity — the model cannot localize information.

Comprehensive Results Summary

Modality	Model	Best Acc.	Paper	Experiments
Images (CIFAR-10)	Perceiver	78.12%	>90%	7 ablations
Images (CIFAR-10)	Perceiver IO	78.20%	—	1
Point Clouds (ModelNet40)	Perceiver	84.24%	85.7%	3 augmentations
Text (WikiText-103)	Perceiver IO	82.20% MLM	—	1
NLU (GLUE, 8 tasks)	Perceiver IO	~63% avg	—	8 fine-tuning

Highlights:

- ModelNet40: only 1.46pp gap from paper
- Perceiver IO slightly edges the best Perceiver
- Successful transfer MLM → GLUE

Overall numbers:

- ~2500 lines of PyTorch code
- 20+ distinct training runs
- 3 modalities: 2D images, 3D points, text
- 100% reproducible via `reproduce.py`

Gap vs Original Paper

ModelNet40 (Point Clouds):

Source	Acc.
Original Paper	85.7%
Our Implementation	84.24%
Gap	1.46pp

Excellent reproduction.

Small gap attributable to reduced batch size (128 vs 512).

CIFAR-10 (Images):

Source	Acc.
Original Paper	>90%
Our Perceiver IO	78.20%
Gap	~12pp

Significant gap.

Caused by hardware constraints: 512 TPU cores vs single consumer GPU.

Root Cause Analysis

The CIFAR-10 gap is primarily due to **batch size** (64 vs 1024), **latent count** (96 vs 256), and **compute time** limitations. The Perceiver architecture is particularly sensitive to these scaling factors on image data.

Hardware Limitations & Design Choices

Parameter	Original Paper (Google)	Our Setup
Hardware	512 TPU v3 cores	1 NVIDIA consumer GPU
Batch size	up to 1024	max 64–128
Latents (N)	256–512	96–128
Model size	30–100M+ params	3.35–10M params
Training time	days (distributed)	hours–days (single GPU)

Consequences:

- LAMB optimizer less effective with small batches
- Fewer latents limit representational capacity
- Reduced augmentation due to time constraints
- Fewer hyperparameter search iterations

Mitigation strategies:

- Weight sharing to reduce memory
- Gradient accumulation where possible
- Careful cosine annealing schedule
- Focus experiments on most informative ablations

Possible Improvements

Architecture:

- Increase latent count ($N = 256+$) with gradient checkpointing
- Add multi-scale cross-attention (coarse $\rightarrow$ fine)
- Explore relative positional encodings (RoPE, ALiBi)
- Deeper decoder for structured outputs

Training:

- Larger batch sizes with gradient accumulation
- Mixed precision (FP16/BF16) training
- Longer training schedules

Data & Tasks:

- ImageNet pre-training for vision
- Subword tokenization for NLP
- Multi-modal joint training
- Audio modality experiments

Analysis:

- Layer-wise attention probing
- Latent space clustering visualization
- Systematic hyperparameter search
- Comparison with efficient Transformers (Linformer, Performer)

Lessons Learned

What works well:

- Genuinely **modality-agnostic** architecture
- Weight sharing: $\sim 2.6\times$ fewer parameters, modest accuracy cost
- Perceiver IO slightly edges base Perceiver on CIFAR-10
- Transfer learning MLM $\rightarrow$ GLUE is effective
- Fourier PE is robust and general-purpose

Observed limitations:

- Performance gap vs paper on images ($\sim 12\text{pp}$)
- LAMB optimizer sensitive to batch size
- Byte-level tokenization produces very long sequences for text
- Augmentation can hurt with limited training
- Hyperparameter tuning is time-consuming

Main Takeaway

Positional encoding is the single most critical component of the architecture: without it, accuracy drops from 72% to 61% on CIFAR-10, confirming the paper's core hypothesis.

Conclusions

Strengths of the Perceiver:

- Single architecture for images, point clouds, and text
- Linear complexity in input size
- Strong memory efficiency via weight sharing ($\sim 61\%$)
- Zero CNN/RNN domain-specific assumptions
- Output queries enable flexible structured decoding

Weaknesses:

- Requires large compute to match specialized models
- Heavily depends on positional encoding
- Outperformed by efficient CNNs on standard vision benchmarks
- LAMB optimizer introduces training instability at small scale
- Latent bottleneck limits fine-grained spatial reasoning

Final Assessment

The Perceiver demonstrates that a **single general-purpose architecture** can handle diverse modalities. While it does not yet match domain-specific models at limited scale, the paradigm

-  A. Jaegle et al., *Perceiver: General Perception with Iterative Attention*, ICML 2021.
-  A. Jaegle et al., *Perceiver IO: A General Architecture for Structured Inputs & Outputs*, ICLR 2022.
-  A. Vaswani et al., *Attention Is All You Need*, NeurIPS 2017.
-  Y. You et al., *Large Batch Optimization for Deep Learning: Training BERT in 76 Minutes*, ICLR 2020.
-  M. Tancik et al., *Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains*, NeurIPS 2020.
-  N. Shazeer, *GLU Variants Improve Transformer*, arXiv 2020.

Appendix: Expected Questions

Architecture:

- ❶ Why cross-attention instead of full self-attention?
→ Reduces $\mathcal{O}(M^2)$ to $\mathcal{O}(NM)$
- ❷ How does weight sharing affect performance?
→ -5.36pp for $\sim 2.6\times$ fewer params
- ❸ Why Fourier PE over learned PE?
→ More robust to permutation, generalizes better
- ❹ What is the role of GEGLU?
→ Higher non-linear capacity, stable gradients

Experiments:

- ❺ Why is the CIFAR-10 gap so large?
→ Hardware limits: batch 64 vs 1024, 96 vs 256 latents
- ❻ Why byte-level tokenization for NLP?
→ Truly modality-agnostic, no external tokenizer
- ❼ How does Perceiver IO improve on Perceiver?
→ Learned output queries vs mean pooling
- ❽ Could this replace ViT/BERT?
→ Not yet at limited scale; promising at Google scale

Thank you!

Questions?

Francesco Gigli e Corso Vignoli

Code and models available upon request

Code:	~2500 lines PyTorch (from-scratch)
Experiments:	20+ training runs, 3 modalities
Reproducibility:	<code>python reproduce.py</code>